# Graphs

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*Solution of the practice quiz*

1. $\mathcal{N} = \{v_0, v_1, v_2, v_3, v_4, v_5\}$ is the set of nodes.
   $\mathcal{A} = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$ is the set of arcs.

2. The incidence function is defined as

$$\Phi : \mathcal{A} \to \mathcal{N} \times \mathcal{N},$$

   and maps the set of arcs to the set of pair of nodes.

$$\begin{array}{ll} \Phi(e_1) = (v_4, v_1)\,, & \Phi(e_2) = (v_4, v_0)\,, \\ \Phi(e_3) = (v_0, v_1)\,, & \Phi(e_4) = (v_3, v_4)\,, \\ \Phi(e_5) = (v_3, v_2)\,, & \Phi(e_6) = (v_2, v_0)\,, \\ \Phi(e_7) = (v_2, v_0)\,. & \end{array}$$

   Note that, in this exampe, the incidence function is not injective because there are two arcs links nodes $v_2$ and $v_0$ ($e_6$ and $e_7$). In the rest of the course, we will only deal with graphs with an injective incidence function.

3. The degree $d_i$ of the node $v_i$ is the number of incident arcs. The out-degree $d_i^+$ of node $i$ represents the number of incident arcs of the form $(i, j)$ (we say that $i$ is an *upstream node*), and the indegree $d_i^-$ of node $i$ is the number of incident arcs of the form $(j, i)$ (we say that $i$ is a *downstream node*).

$$\begin{array}{llllll} d_0 = 4, & d_1 = 2, & d_2 = 3, & d_3 = 2, & d_4 = 3, & d_5 = 0, \\ d_0^- = 3, & d_1^- = 2, & d_2^- = 1, & d_3^- = 0, & d_4^- = 1, & d_5^- = 0, \\ d_0^+ = 1, & d_1^+ = 0, & d_2^+ = 2, & d_3^+ = 2, & d_4^+ = 2, & d_5^+ = 0. \end{array}$$

We notice that for each node $i$ we have $d_i = d_i^- + d_i^+$. Also, we have that $\sum_{i \in \mathcal{N}} d_i^-$ and $\sum_{i \in \mathcal{N}} d_i^+$ are equal to the total number of arcs (7 here).